

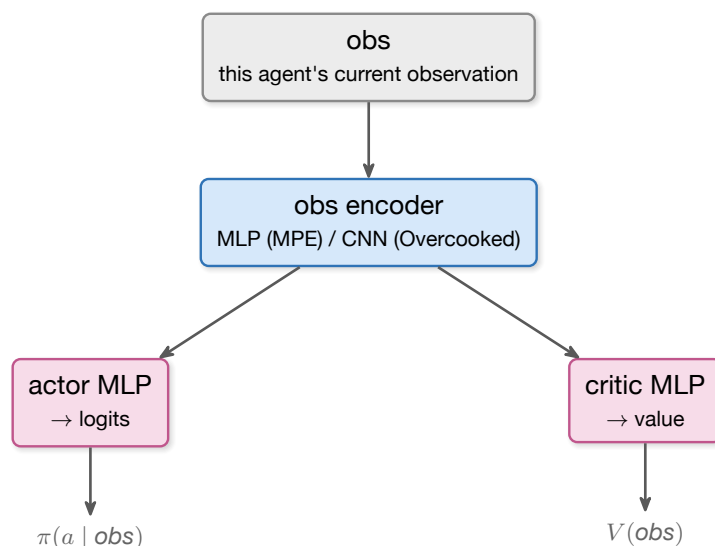
1 • Vanilla IPPO — the baseline

*The control we have to beat. Plain independent PPO: every agent is its own actor–critic, sees only its own observation, and has **no idea it has teammates** beyond whatever leaks into that raw observation.*

In plain English

Each agent runs the *same* small network independently: look at the current observation, encode it, emit (a) an action distribution and (b) a value estimate. No memory, no model of teammates, no auxiliary loss. “IPPO” $\equiv$ “MAPPO” here because Overcooked/MPE agents have full observability, so a shared critic adds nothing. This is a faithful re-implementation of JaxMARL’s validated `ippo_cnn_overcooked` (re-validates at cramped_room $\approx$ 240), so any gain a later architecture shows is real, not a tuning artifact.

Vanilla actor–critic — one observation in, action + value out



Preserved quirk

From the reference: the actor computes a hidden Dense(64) then *ignores it*, taking logits straight off the embedding. It consumes init RNG, so removing it would shift every downstream number — we keep it for exact comparability.

What it does well / badly

- **Well:** simple, stable, strong on near-symmetric tasks where “everyone do the same sensible thing” is already optimal (Overcooked cramped_room, coord_ring — MARC ties here, as expected).
- **Badly — the failure MARC targets:** nothing in the objective or architecture discourages two agents from converging on the *same* role. On MPE spread you literally see agents bunch on one landmark while another sits empty. This is the **redundant / lazy agent** problem.

Intuition: three interns who can each see the whole office but were never told who owns what.

They all gravitate to the obvious task; the awkward one in the corner goes undone.

Results

MPE `simple_spread`, eval return (higher is better; reward is negative). 3 seeds (30/31/32); vanilla is not being expanded — its seed spread is already tiny. These are the numbers every other architecture is measured against.

N	mean	sd	per-seed
3	-19.52	0.12	s30 -19.57, s31 -19.64, s32 -19.36
6	-34.69	0.18	s30 -34.75, s31 -34.88, s32 -34.45
9	-54.03	0.42	s30 -54.59, s31 -53.89, s32 -53.60